DAY 15 — EXCEL BUSINESS ANALYSIS

Topic: PivotTable Fundamentals and Business Performance Analysis

1. LEARNING OBJECTIVES

By the end of Day 15, I learned how to:

- Understand the four main areas of a PivotTable.
- Summarize bookings using Sum, Count, Average, and other calculations.
- Compare countries and products.
- Filter data by year.
- Identify the best-performing country and weakest product by booking volume.
- Analyze monthly trends and investigate performance declines.
- Calculate percentage contribution.
- Drill down into data to identify where further investigation is needed.

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2. PIVOTTABLE FUNDAMENTALS

A PivotTable is an Excel feature used to summarize, organize, compare, and analyze large datasets without changing the original source data.

The Four Main PivotTable Areas

A. ROWS

Purpose: Displays categories vertically.

Examples:

- Country
- Product
- Branch
- Salesperson

B. COLUMNS

Purpose: Displays categories horizontally to make comparisons easier.

Examples:

- Product
- Year
- Quarter
- Status

C. VALUES

Purpose: Performs calculations on numerical or other fields.

Common calculations:

- **Sum:** Adds numerical values.
- **Count:** Counts non-empty records in the selected field.
- **Average:** Calculates the arithmetic mean.
- **Max:** Returns the highest value.
- **Min:** Returns the lowest value.

Important: If the manager asks for total bookings, use Sum of Bookings. If the manager asks how many records exist, Count may be appropriate.

D. FILTERS

Purpose: Restricts the displayed results to a selected category without deleting the original data.

Examples:

- Year = 2025
- Country = Nigeria
- Product = Flight

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3. PRACTICAL TASK 1 — CREATE A PIVOTTABLE

Business Scenario

The manager wants to see total bookings by country, with Hotel and Flight in separate columns, and the ability to filter by year.

Source Dataset

Country	Product	Year	Bookings
Kenya	Hotel	2025	100
Kenya	Flight	2025	150
Kenya	Hotel	2026	200
Kenya	Flight	2026	120
Nigeria	Hotel	2025	180
Nigeria	Flight	2025	90
Nigeria	Hotel	2026	300
Nigeria	Flight	2026	100

Correct PivotTable Field Arrangement

- Rows: Country
- Columns: Product
- Values: Sum of Bookings
- Filters: Year

Steps to Create the PivotTable

1. Ensure the source data has proper column headers.
2. Check that Year and Bookings contain appropriate numeric values.
3. Select any cell within the dataset.
4. Go to Insert → PivotTable.
5. Choose New Worksheet and click OK.
6. Drag Country into Rows.
7. Drag Product into Columns.
8. Drag Bookings into Values.
9. Drag Year into Filters.
10. Verify that the Values calculation is Sum of Bookings, not Count of Bookings.
11. Select the required year using the Year filter.
12. Verify the totals against the source data.

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4. VERIFY PIVOTTABLE RESULTS

Results for 2025

Country	Flight	Hotel	Grand Total
Kenya	150	100	250
Nigeria	90	180	270
Grand Total	240	280	520

Results for 2026

Country	Flight	Hotel	Grand Total
Kenya	120	200	320
Nigeria	100	300	400
Grand Total	220	500	720

Results Across Both Years

Country	Flight	Hotel	Grand Total
Kenya	270	300	570
Nigeria	190	480	670
Grand Total	460	780	1,240

Important: Always verify the year filter before interpreting results. Results for one year should not be confused with totals across all years.

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5. PRACTICAL TASK 1 — BUSINESS ANALYSIS

Question 1: Which country had more bookings in 2026?

Nigeria = 400 bookings
Kenya = 320 bookings

Difference:

400 − 320 = 80 bookings

Conclusion: Nigeria recorded 80 more bookings than Kenya in 2026.

Question 2: Which product had fewer bookings in 2026?

Flight = 220 bookings
Hotel = 500 bookings

Difference:

500 − 220 = 280 bookings

Conclusion: Flight recorded 280 fewer bookings than Hotel in 2026.

Important: Lower booking volume does not necessarily mean lower profitability. Revenue, costs, and profit data are needed to assess financial performance.

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6. PERCENTAGE CONTRIBUTION

Percentage contribution shows how much a particular category contributes to a relevant total.

Formula:

Percentage Contribution = (Category Value ÷ Relevant Total) × 100

Example 1: Kenya's Contribution to Flight Bookings in 2026

Kenya Flight bookings = 120
Total Flight bookings = 220

Calculation:

(120 ÷ 220) × 100 = 54.55%

Conclusion: Kenya contributed approximately 54.55% of Flight bookings in 2026.

Nigeria's contribution:

(100 ÷ 220) × 100 = 45.45%

The two percentages add up to 100%.

Example 2: Nigeria's Contribution to All Bookings in 2026

Nigeria total bookings = 400
Overall bookings = 720

Calculation:

(400 ÷ 720) × 100 = 55.56%

Conclusion: Nigeria contributed approximately 55.56% of all bookings in 2026.

Important: Choose the correct denominator based on the business question.

- Product contribution: Product value ÷ total for the relevant product group.
- Country contribution: Country total ÷ overall total.
- Country-product contribution: Country's value for that product ÷ total for that product.

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7. PRACTICAL TASK 2 — PERFORMANCE INVESTIGATION

Business Scenario

The manager wants to identify the weakest product and investigate why its bookings are lower.

Using the 2026 results:

- Nigeria has 400 bookings.
- Kenya has 320 bookings.
- Hotel has 500 bookings.
- Flight has 220 bookings.

Initial findings:

1. Nigeria has the highest total booking volume.
2. Flight has the lowest total booking volume.
3. Additional analysis is required to understand why Flight bookings are lower.

Monthly Flight Booking Data

Month	Flight Bookings

January	120
February	115
March	118
April	80
May	75
June	78

Trend Analysis

January to March: Bookings remain relatively stable.

April: Bookings drop from 118 to 80.

Calculation:

118 − 80 = 38 fewer bookings.

Percentage decline from March to April:

((118 − 80) ÷ 118) × 100 ≈ 32.20%

May: Bookings fall further to 75.

June: Bookings recover slightly to 78 but remain below March's level.

Conclusion: April is the first major decline to investigate. The data identifies the change but does not explain its cause.

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8. COMPARE PRODUCTS TO UNDERSTAND THE DECLINE

Product	March	April	Change
Flight	118	80	−38
Hotel	200	195	−5

Flight percentage decline:

((118 − 80) ÷ 118) × 100 ≈ 32.20%

Hotel percentage decline:

((200 − 195) ÷ 200) × 100 = 2.50%

Conclusion: Flight has a much larger decline than Hotel. Therefore, Flight-specific factors should be investigated first.

Possible factors include:

- Flight availability
- Pricing changes
- Customer demand
- Marketing campaigns
- Customer enquiries and conversion rates
- Sales processes
- Competitor activity
- Data recording errors

Do not assume a cause without supporting evidence.

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9. COUNTRY-LEVEL DRILL-DOWN

Flight Bookings by Country

Country	March	April	Change
Kenya	70	65	−5
Nigeria	48	15	−33
Grand Total	118	80	−38

Nigeria's decline:

48 − 15 = 33 fewer bookings.

Percentage decline:

((48 − 15) ÷ 48) × 100 = 68.75%

Kenya's decline:

70 − 65 = 5 fewer bookings.

Percentage decline:

((70 − 65) ÷ 70) × 100 ≈ 7.14%

Nigeria's share of the overall decrease:

(33 ÷ 38) × 100 ≈ 86.84%

Conclusion: Nigeria accounts for approximately 86.84% of the overall Flight booking decrease from March to April. It should be prioritized for further investigation.

This identifies where the decline is concentrated, but it does not establish why it occurred.

Recommended Next Investigation

1. Analyze Nigerian branch-level Flight bookings.
2. Compare March and April results by branch.
3. Identify where the largest decline occurred.
4. Investigate salesperson-level data if relevant.
5. Review pricing, availability, customer enquiries, and marketing activity.
6. Verify the source data for completeness and consistency.
7. Recommend corrective action only after gathering sufficient evidence.

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10. THE BUSINESS ANALYST'S INVESTIGATION PROCESS

Follow this sequence when investigating performance problems:

Step 1: Identify the overall performance issue.

Step 2: Compare countries, products, or other relevant categories.

Step 3: Analyze the trend over time.

Step 4: Identify when the significant change occurred.

Step 5: Compare affected and unaffected products.

Step 6: Drill down by country, branch, or salesperson.

Step 7: Investigate possible causes using additional data.

Step 8: Recommend an evidence-based action.

Step 9: Monitor the results after the action.

Key principle: The numbers show what happened and where it happened. Additional evidence helps explain why it happened.

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11. IMPORTANT LESSONS FROM DAY 15

1. Use Rows to display categories vertically.
2. Use Columns to compare categories horizontally.
3. Use Values to calculate sums, counts, averages, and other summaries.
4. Use Filters to focus on selected periods or categories.
5. Use Sum of Bookings when the question asks for total booking values.
6. Verify the results before making business conclusions.
7. Compare actual values and calculate differences.
8. Use percentage contribution to measure a category's share of a relevant total.
9. Investigate trends rather than relying only on grand totals.
10. Drill down into the data to locate the source of a decline.
11. Do not confuse booking volume with revenue or profitability.

12. Avoid assigning blame or recommending actions without supporting evidence.

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12. FINAL REVISION CHECKLIST

Before considering a PivotTable analysis complete, check the following:

☐ Source data has correct headers and data types.

☐ The correct fields are placed in Rows, Columns, Values, and Filters.

☐ The Values field uses the correct calculation.

☐ Year filters are applied correctly.

☐ Grand totals match the source data.

☐ Differences and percentages are calculated correctly.

☐ Business conclusions are supported by the numbers.

☐ Further investigation is identified where the cause is unknown.

☐ Recommendations are based on evidence rather than assumptions.

DAY 15 COMPLETION SUMMARY

Today I learned how to create and verify a PivotTable, compare country and product performance, calculate percentage contributions, identify monthly declines, and drill down into country-level data.

My most important learning is that a business analyst should not stop at identifying a problem. The analyst should investigate the trend, locate where the problem is concentrated, gather evidence about possible causes, and recommend an appropriate action.

Day 15 Completed — PivotTable Fundamentals and Business Performance Analysis.